Phase 4D — Targeted Saturn Transit Presentation & Input-Safety Remediation

✅ **AUDIT STATUS: FULLY REMEDIATED, TESTED & PRODUCTION VERIFIED** **DATE: 2026-10-22**

1,070 / 1,070	Exit Code 0	104 / 135	5 Languages
REGRESSION TESTS (100%)	PRODUCTION VITE BUILD	JPL BENCHMARK (INVARIANT)	EN, HI, KN, MR, TE

1. Executive Summary

Following the **ASTROVEDAM CTO DIRECTIVE — PHASE 4D**, this release safely remediates all four confirmed defects identified during the Phase 4C.3 audit. All existing astronomical algorithms, planetary coordinates, ascendant formulas, and multi-language features were strictly preserved without regression.

2. Remediated Defects & Technical Solutions

DEFECT	ROOT CAUSE	REMEDIATION IMPLEMENTED	STATUS
Fix 1: Sade Sati vs Dhaiya Labels	Kantaka (4th) & Ashtama (8th) Shani rendered as `Active Sade Sati` in reports.	Added distinct localized Dhaiya statuses across all 5 languages (e.g. `Active Dhaiya (Kantaka / Ashtama Shani)` in EN, `शनि ढैया सक्रिय (कंटक / अष्टम)` in HI). Dhaiya is never labeled as Sade Sati.	RESOLVED
Fix 2: Invalid Transit Non-Favorable Contract	Invalid dates & missing Moon signs silently defaulted to `Clear / Favorable (दोष मुक्त)`.	Implemented strict 3-state contract (`valid/affected` , `valid/unaffected` , `invalid/unavailable`). Renders localized slate-gray (`#64748B`) unavailable state; never favorable.	RESOLVED

DEFECT	ROOT CAUSE	REMEDIATION IMPLEMENTED	STATUS
Fix 3: Safe getRashi() Normalization	`getRashi(360.0)` returned out-of-bounds sign `13`, causing runtime `TypeError`.	Safely normalized modulo 360 to `[0, 360)` so zsh^\circ, 360^\circ, 720^\circ ightarrow$ sign 1 (Aries). Negative angles normalize correctly (569X30^\circ ightarrow 12$). Non-numbers & `NaN` return `null` safely.	RESOLVED
Fix 4: Transit Assessment Date Display	Reports did not disclose what assessment date was used, causing confusion with birth date.	Unified single assessment date flow through ephemeris, doshas, predictions, reports, and AstroGuru. Displayed on Page 3 and seeker strip. Date-only parsed at UTC midday.	RESOLVED

3. Regression Test Execution Summary

All 9 regression test suites executed cleanly in **0.39 seconds** with zero errors or warnings:

```
============================================================
📊 REGRESSION TEST EXECUTION SUMMARY
============================================================
✅ PASS | Janampatrika 9-Page Report & Multilingual Integration | 98/98 passed
✅ PASS | North & South Indian Chart SVG Rendering           | 29/29 passed
✅ PASS | Astrology Baseline Contracts & Sade Sati Reference | 27/27 passed
✅ PASS | AstroGuru Conversational AI & Chart Awareness      | 45/45 passed
✅ PASS | Ascendant (Lagna) Astronomical Verification (4B.1)  | 42/42 passed
✅ PASS | Geocentric Planetary Longitude Verification (4B.2)  | 112/112 passed
✅ PASS | Planetary Retrograde & Velocity Verification (4B.3) | 119/119 passed
✅ PASS | Sade Sati & Saturn Transit Dynamic Assessment (4C.2)| 477/477 passed
✅ PASS | Phase 4D Targeted Saturn Transit & Input Safety   | 121/121 passed
------------------------------------------------------------
TOTAL ASSERTIONS: 1,070 | PASSED: 1,070 | FAILED: 0 | DURATION: 0.39s
============================================================
🌟 ALL REGRESSION TESTS PASSED CLEANLY (100% SUCCESS) 🌟
```

4. Remaining Limitations & Astronomical Invariance

- **Saturn Ingress Lag (±2.4 Days):** Left unchanged as mandated by the CTO scope boundaries. The 5-term Meeus perturbation series creates a ~13' orbital error, creating an ingress lag of ~1.8–2.1 days every 2.5 years (e.g. Aquarius ingress on Jan 19, 2023 vs Jan 17, 2023 in JPL). Outside this boundary window, transit sign classification is 100% stable.

- **Benchmark Fixture Provenance:** High-precision geocentric coordinates are verified; original raw web API cURL session logs were not archived.
- **Ascendant & Planetary Invariance:** `calculateAscendant()`, geocentric formulas, retrograde velocities, and Rahu/Ketu routines were 100% untouched.
- **Zero External Dependencies:** 100% local, offline computation preserved.

AstroVedam Astronomical Software Verification Engineering Generated: 2026-10-22 • Version 1.0 • Approved for Release